

RADI-SIMCLIP: A RADIOLOGY VISION-LANGUAGE MODEL PRETRAINED ON SIMULATED RADIOLOGIST LEARNING DATASET FOR ZERO-SHOT MEDICAL IMAGE UNDERSTANDING

Minhui Tan^{1,2,#}, Qingxia Wu^{3,4,#}, Boyang Zhang², Genqiang Ren², Jianlong Nie^{1,2}, Zhong Xue^{1,2}, Xiaohuan Cao^{1,2} ✉, Dinggang Shen^{1,2,5} ✉

¹ School of Biomedical Engineering, ShanghaiTech University, Shanghai, China

² Shanghai United Imaging Intelligence Co., Ltd., Shanghai, China

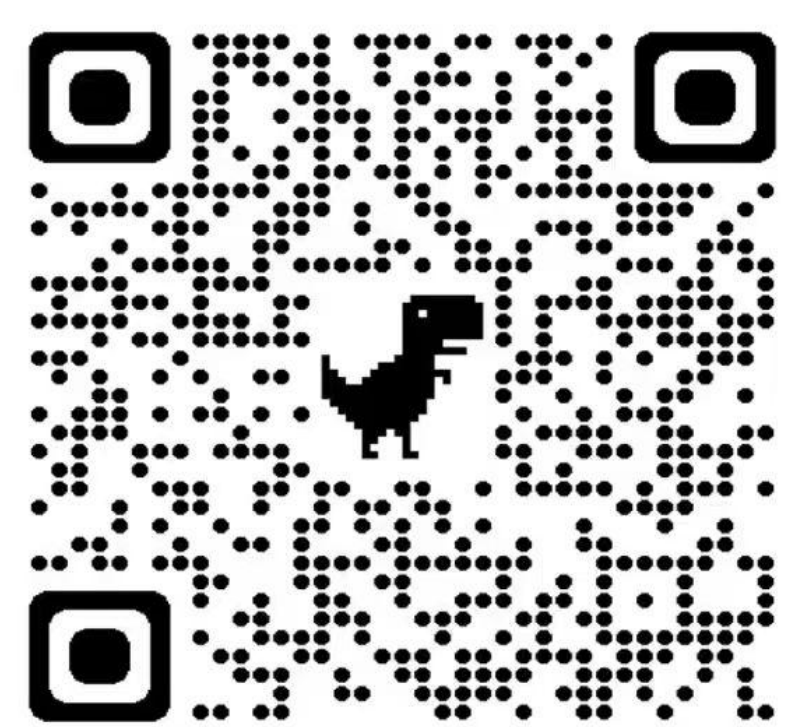
³ Beijing United Imaging Research Institute of Intelligent Imaging, Beijing, China

⁴ United Imaging Intelligence (Beijing) Co., Ltd, Beijing, China

⁵ Shanghai Clinical Research and Trial Center, Shanghai, China

xiaohuan.cao@uii-ai.com Dinggang.Shen@gmail.com

Research Lab
IDEA



INTRODUCTION

Two Limitations of current medical vision-language models:

- **Limited training data:** Lack of radiological diversity and semantic precision.
- **Insufficient evaluation:** Rarely validated on fine-grained tasks.

DATASET

RadiSim

- 10.6M image-text pairs
- Covers **CT**, **MRI**, and **DR** modalities.
- Built from in-house clinical data and curated open-source sources.
- Named RadiSim (“**R**adiology + **S**imulate”) to highlight its goal of reflecting the diverse materials encountered during radiologist training.

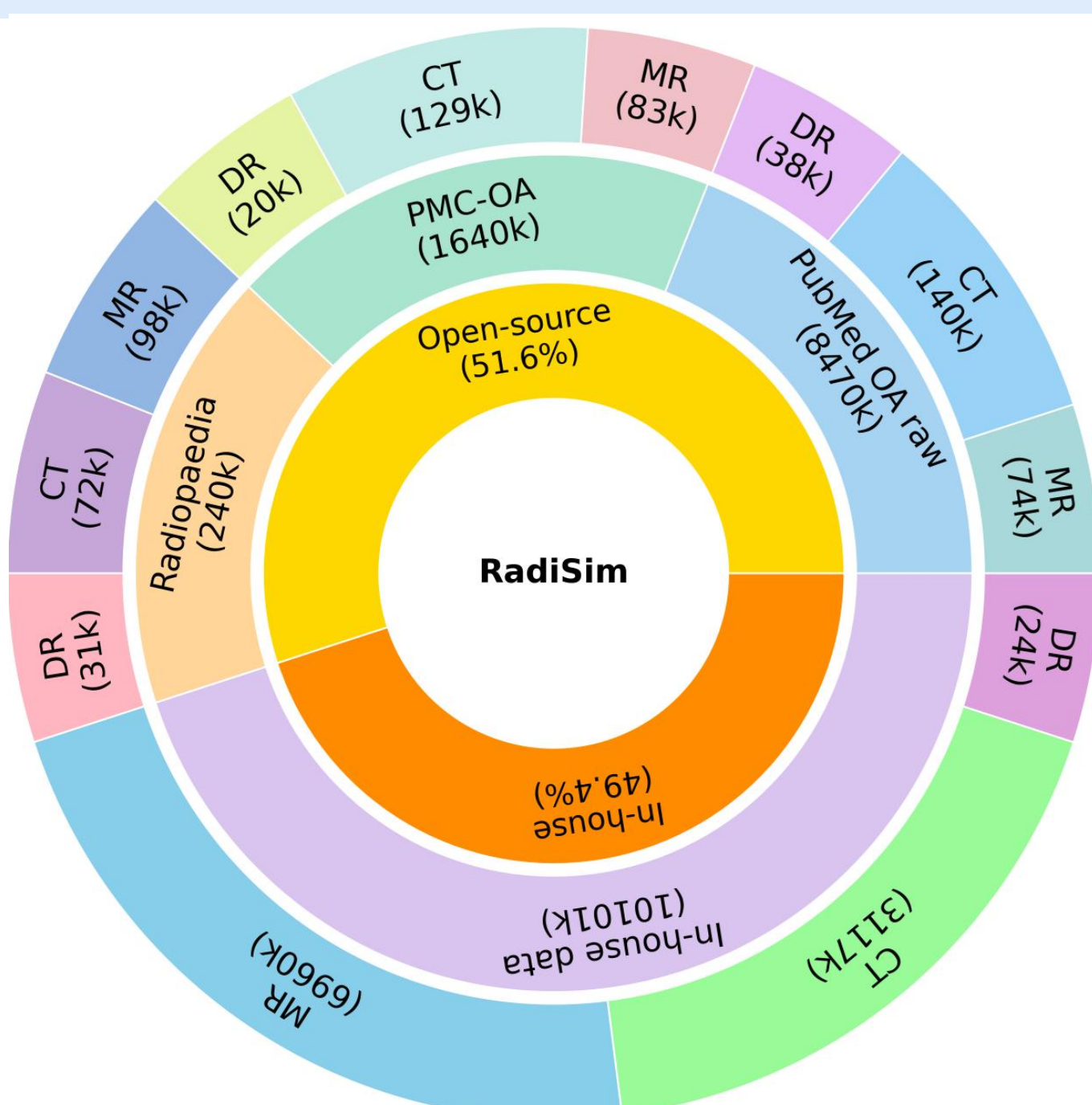


Fig. 1: Overview of RadiSim.

METHOD

Training Strategy: CLIP-based Two-stage Pretraining

- **Stage 1:** Pretraining on public data for **general** vision-language alignment.
- **Stage 2:** **Domain-specific** adaptation on in-house radiology data.

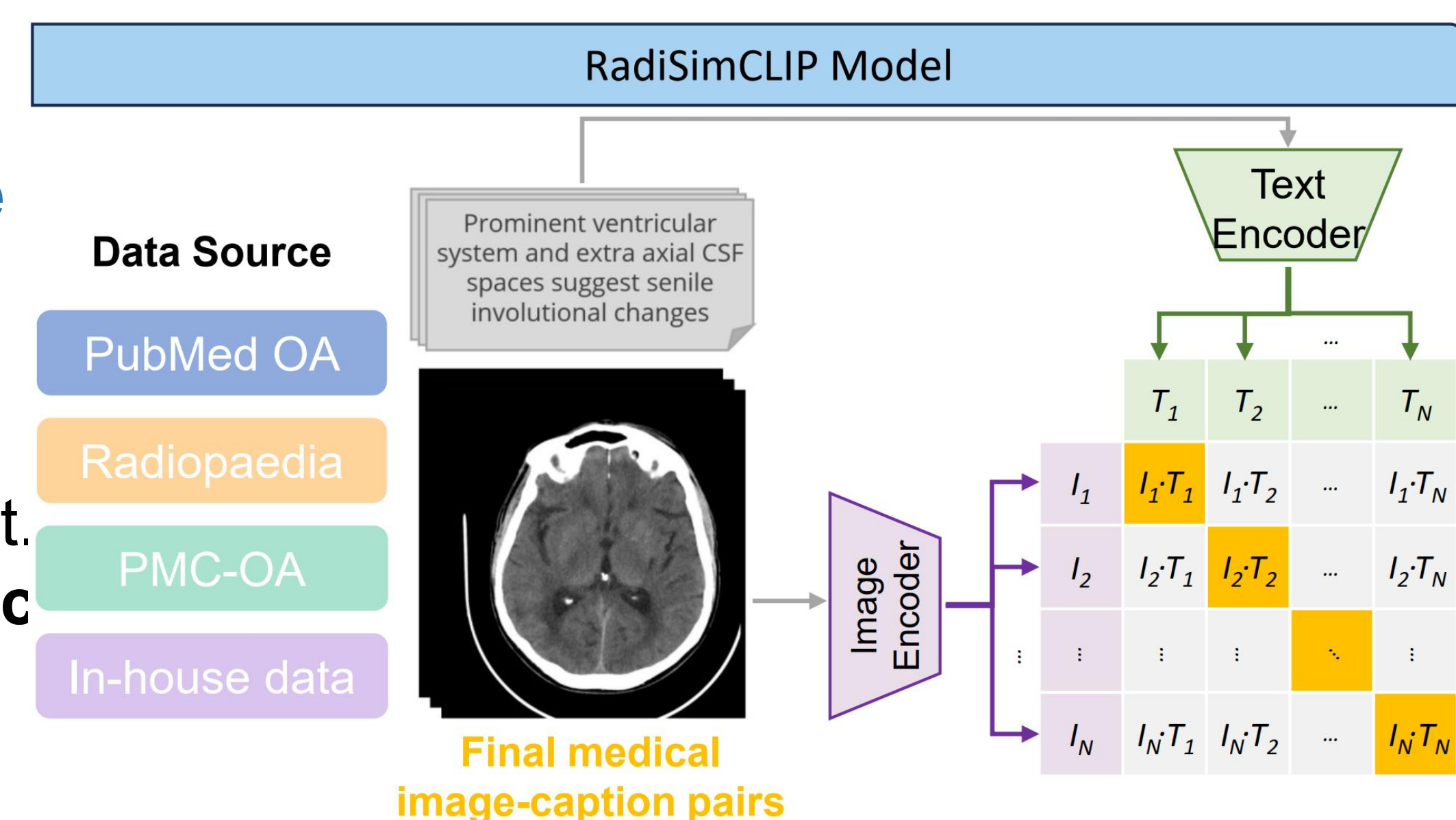


Fig. 2: Overview of RadiSimCLIP training.

EXPERIMENTS

Evaluation Framework: Zero-shot Tasks for Multilevel Capabilities

- **Anatomical, Modality, and Disease Recognition**
 - **2D/3D Zero-shot Classification** via image-text similarity (slice/ volume-level).
- **Spatial Interpretability**
 - **2D Zero-shot Detection** via attention maps to localize organs without fine-tuning.
- **Vision-Language Semantic Alignment**
 - **Cross-modal Retrieval** between images and clinical texts.

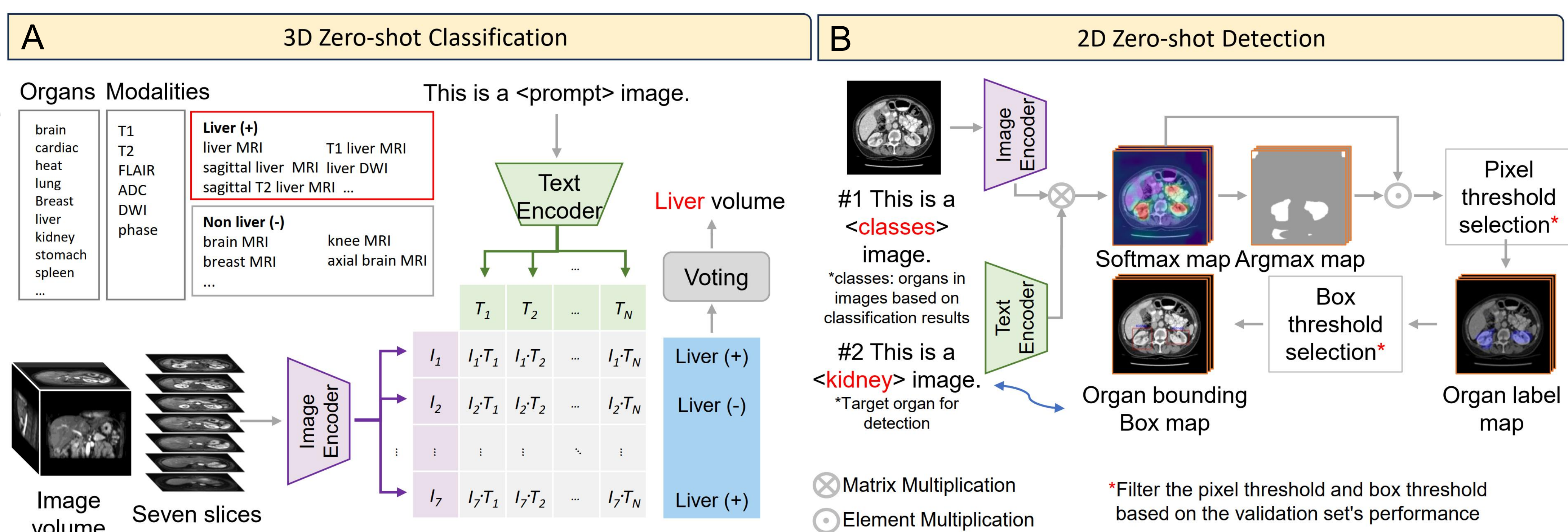


Fig. 3: Two typical downstream tasks: (A) 3D classification, and (B) 2D detection.

RESULTS

- RadiSimCLIP achieves SOTA performance on 15/16 downstream tasks through a multi-level zero-shot evaluation framework.
- Evaluation metrics: Accuracy (ACC), Area Under Curve (AUC), Top-1% accuracy, and Average Rank.

Table 1: Performances of zero-shot classification on different modalities and tasks.

Model	CT (ACC)		DR (AUC)		MRI organ (ACC)		MR modality (ACC)		MRI disease (AUC)	
	Axial	Sag.	Cor.	MNIST	RSNA Brain	Breast	MRA	T1C	ACL	Meniscus
CLIP	0.106	0.106	0.106	0.500	0.500	0.787	0.716	0.155	0.908	0.620
MedCLIP	0.261	0.292	0.314	0.498	0.711	0.763	0.888	0.270	0.852	0.534
PubMedCLIP	0.419	0.343	0.398	0.553	0.646	0.999	0.944	0.864	0.949	0.445
PMC-CLIP	0.300	0.272	0.277	0.556	0.579	0.909	0.803	0.647	0.937	0.618
BiomedCLIP	0.582	0.617	0.644	0.689	0.765	1.000	0.998	1.000	0.998	0.451
Ours	0.765	0.683	0.711	0.728	0.761	1.000	0.999	1.000	1.000	0.707

Table 2: Performance of cross-modal retrieval.

Dataset	BiomedCLIP			Ours		
	Top1%	Acc ↑	Avg. Rank ↓	Top1%	Acc ↑	Avg. Rank ↓
PubMed OA-MR		0.428	2016		0.763	523
PubMed OA-CT		0.698	457		0.781	310
PubMed OA-DR		0.655	111		0.717	97

The best results are in red.

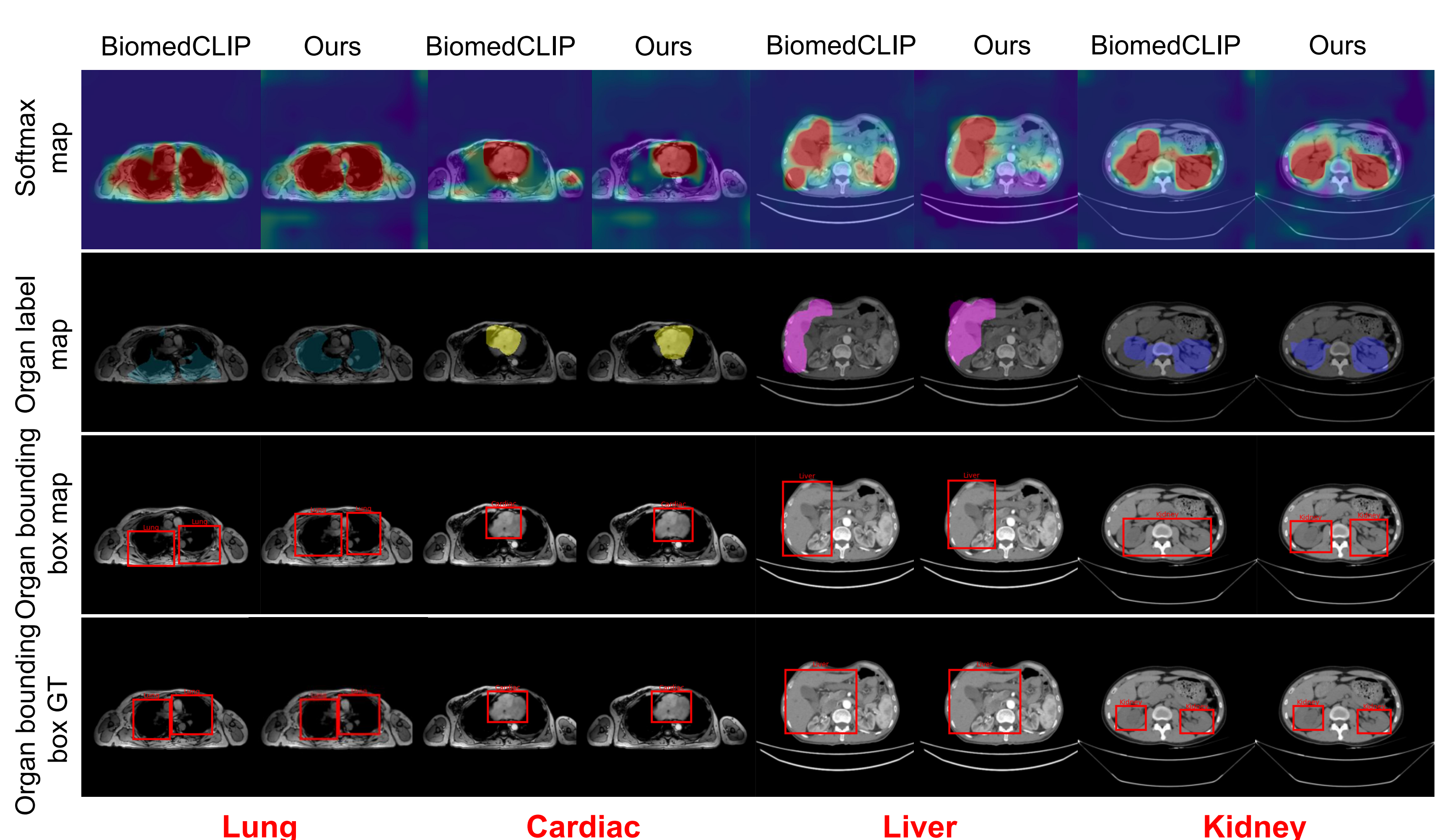


Fig. 4: Results of zero-shot detection across various organs in different datasets.

CONCLUSION

- We have constructed **RadiSim**, a radiology-specific dataset with 10.6M multi-source image-text pairs.
- RadiSimCLIP has achieved SOTA accuracy on **zero-shot understanding** across modalities, including 1) **recognition** via zero-shot classification, 2) **pixel-level understanding** through zero-shot detection, and 3) **semantic alignment** via cross-modal retrieval.